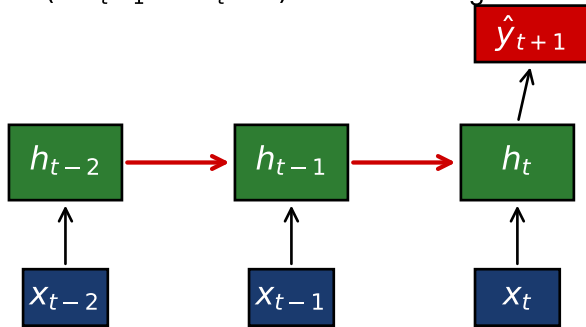
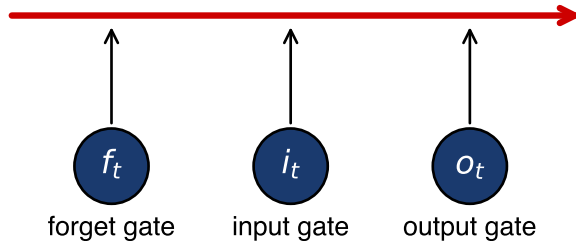


$h_t = \tanh(W h_{t-1} + U x_t + b)$: the same weights at every step



cell state $c_{t-1} \rightarrow c_t$ (long memory)



$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t; \quad h_t = o_t \odot \tanh(c_t)$$